Name _____ Period _____ Date _____

Where Did the Old Ocean Floor Go?

Focus question: How old is the ocean floor compared with the continents, and how do moving plates make and destroy it?

Hook data

Glue in: D1-D Ocean floor and continent ages

Notice and wonder

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
lithosphere (plate)		
plate boundary		
mid-ocean ridge		
subduction		
trench		

Name _____ Period _____ Date _____

Where Did the Old Ocean Floor Go? (continued)

Unit CER: glue U-1 on its own new page and write your first claim.

Plot the pattern

Glue in: your plotted world map. Keep the site list (D1-A) beside it.

Pattern sentence

Plate boundary models

Sketch divergent, convergent, and transform. Label the arrows.

Boundary table

Glue in: D1-B Plate boundary table

Exit ticket

Name _____ Period _____ Date _____

Reading Rock Ages

Focus question: What patterns do the ages of ocean crust and continental crust follow?

Warm-up

Seafloor ages

Glue in: D2-A Seafloor ages and color key

Glue in: D2-B 13-box strip

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
craton		

Name _____ Period _____ Date _____

Reading Rock Ages (continued)

Graph: age vs. distance

Glue in: D2-C Graph grid

Color North America

Glue in: D2-D North America ages and 5-box strip

Add two rows to your Unit CER page (U-1): the ocean pattern and the continent pattern.

Name _____ Period _____ Date _____

Reading Deep Time and Early Earth

Focus question: How do scientists know how old Earth is, and why do meteorites and moon rocks tell Earth's age better than Earth rocks do?

Warm-up

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
impact crater		

Crater counts

Glue in: D3-A Crater counts and three questions

Name _____ Period _____ Date _____

Reading Deep Time and Early Earth (continued)

45 cm Earth timeline

Glue in: D3-B Timeline table and old materials

Your timeline strip

Tape your 45 cm strip here and label each event.

Exit ticket

Add a row to your Unit CER page (U-1) about meteorites, moon rocks, or zircons.

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old

Focus question: How is ocean crust made, grown, and recycled, and how do continents grow and survive?

Warm-up

Guided notes

Glue in: D4-A Guided notes

Sink or float

Glue in: D4-B Density table

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old (continued)

SketchNotes checklist

Glue in: D4-C SketchNotes checklist

SketchNotes: notes

Half a page: the life of ocean crust and how continents grow.

Sensemaking: cross-section

Draw a ridge, moving plate, trench, volcanic arc, and a continent with a younger edge. Label the densities and add a carbon arrow.

Add a row to your Unit CER page (U-1) about density or continental growth.

Name _____ Period _____ Date _____

Evidence for Moving Crust

Focus question: How strong is the evidence that the crust has moved and is still moving?

Warm-up

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
magnetic reversal		

Magnetic stripes

Glue in: D5-A Magnetic stripe band table

Tape in your paper stripes

Tape the stripes here. Label the youngest and oldest crust.

Name _____ Period _____ Date _____

Evidence for Moving Crust (continued)

Hawaii and GPS

Glue in: D5-B Hawaii and GPS table

Pangaea puzzle

Tape your best fit here.

Glue in: D5-C Fossil table

Evaluate the evidence

Glue in: D5-D Evaluate-the-evidence table

Name _____ Period _____ Date _____

Evaluate the Theory, Tell Earth's Story

Focus question: How well does plate tectonics explain the ages of crustal rocks, and what story do ancient materials tell about Earth's formation?

Sort your evidence

Glue in: D6-E Three buckets

Part A: argument

Glue in: D6-A Part A sentence frames

Write your argument

Name _____ Period _____ Date _____

Evaluate the Theory, Tell Earth's Story (continued)

Peer feedback

Glue in: D6-D Star and step

Part B: Earth's story

Glue in: D6-B Evidence cards, in order from oldest to newest

Story frames

Glue in: D6-C Part B story frames

Mark every row of your Unit CER page (U-1) that you used today.

Name _____ Period _____ Date _____

Review Day

Focus question: Which targets do I still need to work on before the CSA?

Self-rating chart

Glue in: D7-A Self-rating chart

Work plan

Name _____ Period _____ Date _____

Review Day (continued)

Vocabulary check

Word	I can explain it to a partner (✓)	My example
lithosphere (plate)		
plate boundary (divergent, convergent, transform)		
mid-ocean ridge		
subduction		
trench		
magnetic reversal		
craton (a continent's ancient core)		
impact crater		

Before the CSA, I will study